Aston University
Birmingham

PROGRAMME SPECIFICATION, UNDERGRADUATE BACHELOR AND LLB

<table>
<tr><td>Programme Title</td><td colspan="6">Electronic Engineering and Computer Science BEng</td></tr>
<tr><td>HECoS Code</td><td colspan="6">100165
100366</td></tr>
<tr><td>College/Subject Area</td><td colspan="6">College of Engineering and Physical Sciences
Electronic and Computer Engineering</td></tr>
<tr><td>Final Award</td><td colspan="6">BEng (Hons.)
BEng (Hons) with Professional Placement for students with placement</td></tr>
<tr><td>Interim Award(s)</td><td colspan="6">Certificate of Higher Education (CertHE)
Diploma of Higher Education (DipHE)</td></tr>
<tr><td rowspan="2">Attendance Pattern</td><td colspan="3">Full Time</td><td colspan="3">Part time</td></tr>
<tr><td colspan="3">X</td><td colspan="3"></td></tr>
<tr><td rowspan="2">Predominant delivery method</td><td colspan="2">Campus-based*</td><td colspan="2">Work-based*</td><td colspan="2">Online/distance</td></tr>
<tr><td colspan="2">X</td><td colspan="2"></td><td colspan="2"></td></tr>
<tr><td>*Location of Study</td><td colspan="6">Aston University campus. The location of placements will be dependent upon the nature of placements available which will be subject to change.</td></tr>
<tr><td>Normal Length of Programme</td><td colspan="6">3 years Full-time, 4 years sandwich</td></tr>
<tr><td>Total Credits</td><td colspan="6">BEng (Hons): 360 credits full-time (plus 120 at level P for sandwich students)
Certificate of Higher Education: 120 credits (at the appropriate level)
Diploma of Higher Education: 240 credits (at the appropriate level)
Ordinary Degree (BSc): 300 credits (at the appropriate level)</td></tr>
<tr><td>Programme Accredited by</td><td colspan="6">The Institution of Engineering and Technology (IET)</td></tr>
<tr><td>Entry Requirements</td><td colspan="6">Entry requirements for each individual student will be stated in their offer letter.</td></tr>
</table>

Educational aims of the Programme	**The Programme aims to:**
	Aim 1: Allow students to develop practical, intellectual and employability skills in the field of electronic engineering and computer science, which will give a competitive edge in the search for placements and graduate employment.
	Aim 2: Use a practical, hands-on approach with engaging projects developing the engineering, problem solving, analysis, management and transferable skills required in graduate employment.
	Aim 3: Make extensive use of project work to develop skills in team working, business and communication.
	Aim 4: Provide students with an opportunity to gain industrial experience in an optional placement year
	Aim 5: Ensure that students are provided with an up-to-date, broad based and balanced study programme through strong links with industry & Aston's internationally recognised researchers

Aston University
Birmingham

Programme Structures and Requirements: Levels, Modules and Credits

Each credit of study is equivalent to 10 learning hours (e.g. 15cr reflects 150 hours of learning). The learning hours may include but are not limited to lectures, seminars, tutorials, lab sessions, practicals, online activity, reading, other independent study, reflecting on assignment feedback, field trips and work placements.

Optional modules are reviewed each year and may change in accordance with the Terms and Conditions of your Contract.

In the table below, a letter P in brackets next to the module code indicates a pre-requisite. A letter C indicates a co-requisite.

STAGE 1

Module Title	Credits	Level	Module Code	Core or Option	Condonable Y/N	Pre- and/or corequisite(s)
Electronics 1	15	4	EI1EL1	Core	N	None
Introductory Mathematics for Engineering, Digital, and Physical Sciences	15	4	EI1IME	Core	Y	None
Introductory Programming for Engineering and Physical Sciences	15	4	DG1IPE	Core	N	None
Interdisciplinary Design Project	15	4	EP1IDP	Core	N	None
Foundations of AI and Data Science	15	4	DG1AID	Core	Y	None
Electronics 2	15	4	EI1EL2	Core	Y	(C) EI1EL1
Internet Applications and Databases	15	4	DG1IAD	Core	Y	None
Power Skills	15	4	EP1POS	Core	N	None
TOTAL	120					

STAGE 2						
Module Title	**Credits**	**Level**	**Module Code**	**Core or Option**	**Condonable Y/N**	**Pre- and/or corequisite(s)**
Analogue and Power Electronics	15	5	EI2APE	Core	Y	None
Robot Control	15	5	EI2ROC	Core	Y	(C) EI2ESC
Data Structures, Algorithms and Object-Oriented Programming	15	5	DG2OOP	Core	Y	(P) DG1IPE
Electronic Engineering Team Project	15	5	EI2ETP	Core	N	None
Communications Systems Ethics, EDI and Sustainability	15	5	EI2COS	Core	N	None
Embedded Systems and C	15	5	EI2ESC	Core	N	None
Digital Design	15	5	EI2DID	Core	N	None
Introduction to Artificial Intelligence and Robotics	15	5	DG2IAI	Core	Y	None
TOTAL	120					

STAGE 3P (Placement, short placement modules or study abroad)						
Module title	**Credits**	**Level**	**Module Code**	**Core or Option**	**Condonable Y/N**	**Pre-requisite(s)**
Students may select the following option:						
EPS Placement Year	120	P	EPSP01	Option	N	None
TOTAL	120					

STAGE F (Final)						
Module Title	**Credits**	**Level**	**Module Code**	**Core or Option**	**Condonable Y/N**	**Pre- and/or corequisite(s)**
Internet of Things	15	6	EI3IOT	Core	Y	None
Digital Systems Design	15	6	EI3DSD	Core	Y	(P) EI2DID
Professional Engineering Practice	15	6	EI3PEP	Core	N	None
Final Year Project	45	6	EI3EFP	Core	N	None
Choose 15 credits from the each of the following sections						
Section A						
Digital Signal Processing	15	6	EI3DSP	Option		None
Autonomous Robotics and Navigation	15	6	DG3ARN	Option	Y	(P) EI2ROC
Computer Animation	15	6	DG3CAN	Option	Y	None
Software Testing	15	6	DG3SFT	Option	Y	None
Section B						
Digital Systems Hardware	15	6	EI3DSH	Option	Y	(P) EI2DID
Adaptive and Intelligent Robotic Systems	15	6	DG3ARS	Option	Y	None
Data Mining	15	6	DG3DAM	Option	Y	None
Cloud Computing and DevOps	15	6	DG3CCD	Option	Y	None
Game Development	15	6	DG3GAD	Option	Y	None
TOTAL	120					

Aston University
Birmingham

Programme learning outcomes

Achievement of programme learning outcomes is demonstrated through module assessment at the appropriate level/year of study.

Stage 1 (First Year, should map to FHEQ Level 4 equivalent to an interim award of Cert.HE)	
	On successful completion of this level, students will:
LO4.1	Be able to apply knowledge of circuit theory and programming principles to design basic electronic & computing systems
LO4.2	Have demonstrated knowledge of and the ability to apply computing and physics principles in engineering contexts
LO4.3	Be able to identify, evaluate and apply mathematical, programming, and computational approaches tosolving complex problems
LO4.4	Be able to analyse measurement results and interpret technical data sheet specifications to make reasoned engineering design decisions
LO4.5	Have demonstrated the ability to effectively use electronic engineering components, test equipment and design, analysis and programming software
LO4.6	Be able to effectively report and communicate technical results and independent study both in writing and orally
LO4.7	Have developed self-study skills and independent learning skills to facilitate professional development
LO4.8	Have developed an appreciation of the team-working and project management skills required in professional engineering employment
LO4.9	Evaluate and interpret data using industry-standard methods and communicate information effectively to ensure data-driven approaches to decision-making in solving complex problems.
LO4.10	Use sustainable development strategies that reflect a deep understanding of the Circular Economy, Net-Zero strategies, Design and Systems Thinking, and Global Social Responsibility and lead to solutions that are ethically sound and globally relevant.
LO4.11	Employ integrated problem-solving skills to analyse challenging and multifaceted problems considering professional, social, and ethical contexts.

Aston University
Birmingham

Stage 2 (Second Year, should map to FHEQ Level 5 equivalent to an interim award of Dip.HE)	
	On successful completion of this level, students will:
LO5.1	Have demonstrated a critical understanding of the principles of a broad range of topics in the field of electronic and software engineering and their relevance to contemporary systems and applications
LO5.2	Be able to apply an understanding of mathematical, programming and engineering concepts and analysis techniques to develop solutions to a wide range of electronic engineering and computing problems
LO5.3	Have demonstrated the ability to design, evaluate and critically review solutions to an engineering problem subject to technical, commercial and societal constraints
LO5.4	Be able to undertake independent critical analysis of information from a range of sources, evaluate and interpret it to provide valid, reasoned technical recommendations and arguments
LO5.5	Have the ability to identify and apply the practical lab and computer aided design skills required for the development of new engineering designs and solutions
LO5.6	Be able to apply programming and computing skills to the structured design, modelling and implementation of engineering systems
LO5.7	Be able to communicate both technical and non-technical information at appropriate levels for different audiences both in writing and orally
LO5.8	Have demonstrated the ability to apply industry standard team working and project monitoring skills in the development of a prototype electronic and/or computing system

Stage 3 (Placement Year, Level P (not mapped to FHEQ))	
	On successful completion of this level, students will:
LOP.1	Develop new knowledge and understanding of a commercial engineering environment, the role of individuals within project teams and associated professional engineering issues
LOP.2	Be able to communicate effectively in a professional engineering environment
LOP.3	Reflect on experience studying in a different culture leading to personal development in global awareness, intercultural competency and adaptability

Stage F (Final Year, should map to FHEQ Level 6 and a final award of BEng Hons)	
	On successful completion of their programme, students will be able to:
LO6.1	Demonstrate a systematic understanding of established knowledge and practice in electronic engineering and computer science and an in depth knowledge of current developments in a specialist area
LO6.2	Research, review and evaluate information from a range of sources to develop knowledge and understanding of advanced concepts
LO6.3	Apply a conceptual understanding to evaluate different problem solving approaches and methodologies and to identify new approaches to the solution
LO6.4	Use engineering judgement to recognise and adapt information which may be partially specified and resolve conflicting inter-disciplinary requirements
LO6.5	Adopt appropriate investigative techniques to devise and critically evaluate solutions to an engineering problem subject to multiple real-world constraints
LO6.6	Select and use appropriate communication styles and formats to convey information and ideas at an appropriate level for academic, technical and nontechnical audiences
LO6.7	To evaluate, reflect on and communicate their personal strengths and limitations; to appreciate the implications of these and be able to identify appropriate professional development opportunities
LO6.8	Plan, monitor and implement a substantial individual contribution to an engineering project to successfully achieve a measurable set of objectives

Assessment Types
The programme will be assessed through a combination of written and oral examinations, class tests, individual and group coursework, projects, presentations and practical assessments.

Aston University
Birmingham

Approved Exemptions from General Regulations

In order to gain exemption from the IET examination requirements for progression towards Chartered Engineer status the following conditions must be satisfied: - Where a module comprises two or more separate assessments, in order to pass the module students must achieve a mark of at least 30% in each separate assessment which is worth more than 30% of the overall final mark of the module (as listed in the module specification). This is in addition to the overall module mark being at or above the pass mark (40%) in order to achieve a pass for the module as a whole. General Regulations will apply to the reassessment of failed components. - In case of having mark below 30% in the assessment worth more than 30% (component fail), but overall module mark at pass level and student is meeting all the learning outcomes of the programme in other modules, it would be up to the Board of Examiners discretion to waive this rule to pass the module for the student. - It is a requirement of the IET that no more than 30 credits can be compensated (condoned) in a Bachelors (BEng) or integrated Masters (MEng) degree programme.

General Regulations (https://www2.aston.ac.uk/clipp/quality/a-z/general-regulations) and the Regulations for the programme (above) take precedence over other information sources such as student handbooks if there is a conflict. If there is a conflict between General Regulations and Programme Regulations then General Regulations take precedence unless an exemption has been approved.

For administrative use only:

Dates Programme Specification Written and Revised	Written: August 2010 Revised: September 2014, July 2015, 1 August 2015, 1 April 2016, 10 May 2016, 16 August 2016, 28 November 2016, 2 June 2017, April 2018, 15 August 2018, June 2019, August 2019, September 2019, February 2020, May 2021, July 2021, August 2022, December 2022, May 2024, July 2024, March 2025, August 2025